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Statistical Analysis Report

World Cities, Weather Forecasts & Bike-Sharing Systems

Source: SQL EDA query output dump (124 queries across 4 base tables) **Report date:** September 17, 2026

How to read this report

This report is built entirely from a dump of *pre-computed* SQL query results (aggregates, rankings, and joins), not from the raw underlying rows. That has two consequences:

1. Wherever the source data contains a **NULL / NA / missing value**, this report simply flags it and notes that **it will be addressed in the next DATA SCIENCE phase** — no imputation or guessing is done here.
2. Deeper statistical work that requires raw records (correlation coefficients, regression, significance testing, outlier removal) is out of scope for this pass and is recommended as follow-up work in that same phase.

Four underlying tables were queried:

Table	Rows	Description
WORLD_CITIES	26,569	Global gazetteer of cities: country, ISO codes, coordinates, population
WEATHER_FORECAST	160	Short-range (3-hour interval) forecast for 4 cities, Sep 14-19, 2026
SEOUL_BIKE_SHARING	8,465	Hourly bike-rental demand for Seoul, Dec 2017-Nov 2018
BIKE_SHARING_SYSTEMS	480	Global directory of bike-share systems by city/country with fleet size

Across these, 124 distinct queries were executed, including several **cross-table joins** (WORLD_CITIES ↔ BIKE_SHARING_SYSTEMS, and a 3-way join with WEATHER_FORECAST). A recurring

and important theme of this report is that several of those joins have a **data-quality problem (fan-out from non-unique city names)** that materially distorts some of the derived metrics — this is documented in detail in Section 7.

1. Executive Summary

- The world-cities reference table covers **26,569 cities in 224 countries**, but has **973 missing population values (3.7%)**, **1,073 duplicated CITY+COUNTRY combinations** (accounting for ~1,881 redundant rows, ~7.1% of the table), and several implausible population entries (e.g., a city listed at a population of “4.472”).
- Population is heavily right-skewed: **mean 162,346 vs. median 20,080** — a handful of megacities (Tokyo, 38.0M; Jakarta, 34.5M; Delhi, 29.6M) pull the average far above the typical city.
- The Seoul bike-share system shows classic commuter-driven demand: two daily peaks, with **18:00 the single busiest hour (avg. 1,554 rentals)**, a **~4.6× swing between summer and winter demand**, and strong, non-linear sensitivity to rain, snow, and humidity (demand drops ~70-78% in the presence of rain or snow).
- The global bike-share systems directory covers **87 systems in 434 cities across 57 countries**, totaling roughly **813,000 bicycles** across countries with known fleet sizes — of which **China alone accounts for ~58%** (469,941 bikes). **15.8% of system records are missing a fleet-size value.**
- The join between WORLD_CITIES and BIKE_SHARING_SYSTEMS only cleanly matches **271 of 434 bike-share cities (62%)**; the remaining **163 (38%) fail to match**, and worse, **57 cities that do match produce multiple rows** — because the join appears to key on city name alone, without country. This causes some very large, misleading results (e.g., an implied population for “Paris” of 1.85 million, derived by averaging the real Paris, France against five small U.S. towns also named Paris). **All per-capita bike-share figures downstream of this join should be treated with caution.**
- Several individual queries in the dump appear to have **logic issues** unrelated to missing data (e.g., a “sort by highest population” query that returns the same order as the unsorted version; a “largest bike-share systems” query that appears to count records rather than sum fleet size). These are flagged inline.

2. World Cities Dataset

2.1 Coverage & Scale

Metric	Value
Total cities	26,569
Unique countries	224
Unique ISO2 codes	223
Unique ISO3 codes	224
Cities with missing population (NA)	973 (3.7%)
Cities with population = 0	5

Data quality flag: ISO2 codes (223) don’t line up with the country count (224) or ISO3 codes (224) — one country appears to have a missing or duplicated ISO2 value. **To be addressed in the next DATA SCIENCE phase. Data quality flag:** The 973 missing population values are NA and are not imputed or estimated anywhere in this report. **To be addressed in the next DATA SCIENCE phase.**

2.2 Duplicate CITY + COUNTRY Records

The dataset contains **1,073 CITY+COUNTRY combinations that appear more than once** — together responsible for **~1,881 redundant rows ($\approx 7.1\%$ of all 26,569 cities)**.

# of duplicate rows for a combination	# of combinations affected
2	709
3	186
4	72
5	42
6	23
7	21
8	4
9	8
10	3
11	2
12	1
13	1
14	1

This is **overwhelmingly a U.S. data issue**: of the 1,073 duplicated combinations, **890 (83%)** are U.S. cities (e.g., “Ashland”, “Auburn,” and “Burlington” each appear under 9 different U.S. records — almost certainly same-named towns in different states that weren’t disambiguated at the city+country grain). The remaining duplicates cluster in China (31), Japan (28), Mexico (19), Russia (18), and Germany (12).

This same-city-name issue re-appears later as the root cause of the join fan-out problem in Section 7 — it is one of the most consequential data-quality issues in the whole dump.

2.3 Population Distribution

Metric	Value
Average population	162,345.7
Median population	20,079.5
Cities > 100,000	4,522 (17.0%)
Cities > 1,000,000	714 (2.7%)
Cities > 5,000,000	145 (0.5%)

The mean is **$\sim 8\times$ the median**, confirming a classic long-tailed population distribution: most cities are small, and a small number of very large cities dominate the sum.

Largest 10 cities:

Rank	City	Population
1	Tokyo	37,977,000
2	Jakarta	34,540,000
3	Delhi	29,628,735
4	Mumbai	23,355,000
5	Manila	23,088,000

Rank	City	Population
6	Shanghai	22,120,000
7	São Paulo	22,046,000
8	Seoul	21,794,000
9	Mexico City	20,996,000
10	Guangzhou	20,902,000

Smallest non-zero cities: the single smallest is **Kobarid at a population of “4.472”** — almost certainly a units/scale error (a fractional population under 5 people is not realistic for a named city). Nine more cities tie at exactly **10**, a suspiciously round number that looks like a placeholder/floor value rather than a measured population. **These are not NA, but they are flagged as likely data-entry or unit errors for review in the next phase.**

2.4 Geographic Distribution

Metric	Value
Northernmost city	Nord, Greenland (81.72°N)
Southernmost city	Puerto Williams, Chile (54.93°S)
Easternmost city	Labasa, Fiji (179.37°E)
Westernmost city	Zvëzdnny, Russia (179.59°W — i.e., just west of the anti-meridian)
Cities north of equator	23,797 (89.6%)
Cities south of equator	2,772 (10.4%)

The dataset is heavily skewed toward the Northern Hemisphere (~9:1), consistent with the global distribution of land and population, not a data artifact. Cities closest to the equator include Ntara, Uganda (0.0044°) and Makoua, Republic of the Congo (−0.0047°).

2.5 Country-Level Breakdown

Top 10 countries by number of cities:

Country	City count
United States	7,768
United Kingdom	1,798
Germany	1,606
Russia	1,161
Japan	827
Brazil	788
Mexico	750
China	693
Canada	625
France	444

The U.S. alone contributes **29% of all cities in the table** — combined with the U.S. also contributing 83% of duplicate records, this suggests the U.S. slice of this gazetteer was sourced or processed differently (e.g., town/township-level granularity) than most other countries.

Total population represented by cities, top 10 countries (sum of city-level populations, i.e. not necessarily the true national population where cities/records are incomplete or duplicated):

Country	Summed city population
China	1,388,868,247
United States	400,521,452
India	270,170,371
Japan	148,273,773
Brazil	135,416,434
Russia	105,990,872
Mexico	103,464,677
Indonesia	85,283,563
United Kingdom	72,555,282
Turkey	70,958,511

China’s figure (1.39B) lines up closely with its real-world population, which is a good sanity check on the data overall. The U.S. figure (400.5M) is notably **higher than the real U.S. population (~335M)** — plausibly inflated by the large number of duplicated U.S. city records identified in Section 2.2, which would double-count population for any duplicated town.

Query-logic flag: A separate query, “*Find countries whose largest city has the highest population*” (intended to rank countries by their largest city’s population), returned output that is **byte-for-byte identical** to the unranked “*largest city in each country*” query. The sort/ranking step does not appear to have taken effect — this output should not be read as a ranking.

Data quality flag: The “*largest city per country*” aggregate only returns **221 of the 224 countries** in the dataset. The 3 missing countries most likely have city records where population is entirely NA, so no city can be selected as the country’s “largest.” **To be addressed in the next DATA SCIENCE phase.**

3. Weather Forecast Dataset

3.1 Overview

Metric	Value
Total records	160
Unique cities	4 (Paris, Seoul, Suzhou, Washington, D.C.)
Unique weather conditions	3
Forecast window	2026-09-14 18:00 → 2026-09-19 15:00
Records per city	40 each

Forty 3-hour-interval records per city over roughly 5 days ($40 \times 3\text{h} \approx 120\text{h}$) is consistent with a standard “5-day / 3-hour-step” forecast product.

3.2 Temperature, Humidity, Wind & Pressure by City

City	Avg. Temp (°C)	Avg. Humidity (%)	Avg. Wind Speed	Avg. Pressure (hPa)
Suzhou	23.68	71.10	2.31	1,017.3
Seoul	22.71	44.53	1.91	1,019.7
Paris	19.55	51.95	2.61	1,018.7
Washington, D.C.	16.62	56.10	1.78	1,017.0

- **Warmest single reading:** Paris, 33.83 °C (2026-09-15 15:00)
- **Coolest single reading:** Paris, 10.02 °C (2026-09-16 06:00)

Notably, **the warmest and coolest readings in the entire dataset both belong to Paris**, roughly 15 hours apart — a **~24 °C swing**, which is a very large single-city range for a mid-September forecast and worth double-checking against the raw feed in the next phase.

Query-logic flag: A separate query for “*city with largest temperature amplitude*” reports **Washington, D.C. at only 3.09 °C** — far smaller than Paris’s actual 23.81 °C swing described above. This strongly suggests the “amplitude” query computed something narrower (e.g., a single record’s TEMP_MAX - TEMP_MIN) rather than the city’s full range across the forecast window, and should not be read as the true highest-amplitude city.

Pressure is tightly clustered (1,017–1,020 hPa) across all four cities, consistent with normal, unremarkable atmospheric conditions — no severe weather is implied by pressure alone.

3.3 Weather Conditions

- Most common condition by city: **Paris — Clouds (34/40, 85%), Suzhou — Clouds (31/40, 78%), Washington, D.C. — Clear (26/40, 65%), Seoul — Clouds (22/40, 55%).**

Data quality flag: The “*count weather conditions by season*” query labels **all 160 records as “Winter,”** regardless of city or the fact the forecast window is mid-September (early autumn in the Northern Hemisphere, early spring in the Southern Hemisphere). This is very likely a season-assignment bug (e.g., a lookup hard-coded to Seoul’s climate definitions applied globally) rather than a real finding. **To be addressed in the next DATA SCIENCE phase.**

Query-logic flag: The query for “*the weather forecast for Seoul over the next 3 hours*” returned only **1 row** instead of the expected 3, and that row’s timestamp is displayed as a raw Unix epoch integer (1789408800) rather than a formatted datetime — which converts to **2026-09-14 18:00:00**, i.e., the very first record in the whole dataset, not a “next 3 hours” window relative to any current time. Both the row count and the field formatting look like defects in that query.

Query-logic flag: “*Find the largest city under each weather condition*” returns a malformed result — its WEATHER_CONDITION column contains values like “0.43 (WIND_SPEED)”, “15.88 (TEMPERATURE)”, and “24.0 (HUMIDITY)” instead of actual condition labels (e.g., Clear/Clouds/Rain). This output does not answer the intended question and should be re-run.

4. Seoul Bike-Sharing Dataset

4.1 Overview & Completeness

Metric	Value
Total records	8,465
Date range	2017-12-01 → 2018-11-30
Unique dates	353
Seasons present	Winter, Spring, Summer, Autumn

The date range spans 365 calendar days, but only **353 unique dates** are present — **12 calendar dates are missing entirely** from the dataset ($365 - 353 = 12$), and total records (8,465) fall short of a full $24 \times 365 = 8,760$, a shortfall of 295 hours. This is broadly consistent with 12 fully-missing days (288 hours) plus a handful of additional missing hours on otherwise-present days. **This is a completeness gap, not a NULL value per se, but is noted for the next phase.**

Records per season: **Autumn 1,937, Spring 2,160, Summer 2,208, Winter 2,160** (sums to 8,465). Autumn’s shortfall relative to the other seasons (~220 fewer records) is consistent with the missing dates being concentrated there.

Note: The query “*how many hours had non-zero rented bike count*” returns exactly 8,465 — identical to the total row count. Combined with the dataset minimum of 2 rentals/hour, this confirms **every recorded hour had at least some rentals**; there are no true zero-demand hours in this dataset.

4.2 Overall Demand Statistics

Metric	Value
Average rentals/hour	729.16
Minimum rentals/hour	2
Maximum rentals/hour	3,556
Peak record	2018-06-19, 18:00 (3,556 rentals)
Lowest records (tied at 2)	2018-03-05 03:00; 2018-04-23 04:00; 2018-10-06 10:00

4.3 Hourly Demand Patterns

Hour	Avg. rentals	Hour	Avg. rentals
00	561	12	723
01	442	13	758
02	313	14	785
03	211	15	857
04	137	16	962
05	144	17	1,177
06	298	18	1,554
07	627	19	1,236
08	1,050	20	1,105
09	668	21	1,067
10	546	22	954
11	621	23	694

This is a textbook **bimodal commuter pattern**: a trough around 04:00–05:00 (~137–144 avg. rentals), a morning shoulder peak at 08:00 (1,050), and a much larger, broader evening

peak from 17:00–21:00, topping out at **18:00 (1,554 avg. rentals — the single busiest hour of the day)**.

5 busiest hours: 18:00 (1,554), 19:00 (1,236), 17:00 (1,177), 20:00 (1,105), 21:00 (1,067) — notably, **all five are evening hours**; the morning peak (08:00, 1,050) doesn’t crack the top 5, indicating the evening commute/leisure window drives materially more demand than the morning commute.

5 quietest hours: 04:00 (137), 05:00 (144), 03:00 (211), 06:00 (298), 02:00 (313).

Daytime vs. nighttime totals: 3,202,422 vs. 2,969,892 — a roughly even split, tilted slightly toward daytime (~52%/48%).

The full hour-by-season breakdown (96 hour×season combinations, with mean, min, max, and standard deviation) confirms the same 18:00 peak recurs in **every season**, though at very different scales — e.g., the Summer 18:00 average (2,135) is almost **9× the Winter 18:00 average** (438).

4.4 Seasonal & Monthly Patterns

Season	Avg. rentals	Median	Max	Min
Summer	1,034.07	905.5	3,556	9
Autumn	924.11	856.0	3,298	2
Spring	746.25	599.0	3,251	2
Winter	225.54	203.0	937	3

Summer demand is **~4.6× Winter demand** on average — the single strongest driver of variance in this dataset.

Monthly totals (Dec 2017 – Nov 2018): the **busiest month is June (896,887 total rentals, avg. 1,246/hr)**; the **quietest is January (150,006 total, avg. 202/hr)** — over a **6× gap** between the busiest and quietest months.

Seasonal weather averages line up with the demand pattern: Summer averages 26.6 °C / 65% humidity, while Winter averages **−2.5 °C** / 50% humidity — with correspondingly minimal solar radiation (0.30 vs. Summer’s 0.76) and the only meaningful average snowfall (0.25mm, vs. 0 in Summer/Spring).

Query-logic flag: “*Find the busiest and quietest individual date*” returns the same single-hour extreme values already seen above (3,556 on 2018-06-19; the three tied “2” observations), rather than daily *totals*. A true “busiest date” query should sum rentals across all 24 hours of a date before ranking — this result should be re-run as a proper daily aggregate rather than reused as-is. Similarly, “*busiest/quietest hour within each season*” surfaces the same handful of globally extreme records (value = 2) rather than a season-specific ranking, since ties at the dataset minimum happen to fall in different season/hour buckets (Autumn/10:00, Spring/03:00, Spring/04:00). This should also be re-verified.

4.5 Weather Sensitivity

Temperature:

Temp. range	Records	Avg. demand	Max demand
20–30 °C	2,340	1,148.13	3,556

Temp. range	Records	Avg. demand	Max demand
30 °C+	524	1,112.25	3,238
10-20 °C	2,050	837.16	2,857
0-10 °C	2,118	426.34	2,202
< 0 °C	1,433	197.99	1,671

Demand rises steadily with temperature up to the 20-30 °C band, then **plateaus/slightly dips above 30 °C** — a classic inverted-U relationship where extreme heat, like extreme cold, modestly suppresses ridership relative to the comfortable middle band.

Rainfall:

Rain group	Records	Avg. demand	Max demand
No rain	7,949	765.63	3,556
>0-1 mm	274	224.00	2,598
>1-5 mm	178	113.70	1,208
>5 mm	64	73.28	858

Even a trace of rain (>0-1mm) is associated with a **~71% drop** in average demand versus no rain — the effect is front-loaded rather than gradual. Overall, records with any rain (n=516) average 167.26 rentals/hr vs. 765.63 without (a 78% reduction).

Snowfall:

Snow group	Records	Avg. demand	Max demand
No snow	8,022	759.20	3,556
>0-1	255	186.57	1,524
>1-3	134	199.34	846
>3	54	142.85	436

Records with any snow (n=443) average 185.10 rentals/hr vs. 759.20 without — a **76% reduction**, similar in magnitude to rain. Note the >0-1 and >1-3 buckets are not perfectly monotonic (186.57 vs. 199.34) — likely sample-size noise in the smaller buckets rather than a real effect, worth re-checking with more data in the next phase.

A seasonal cut shows this effect holds even in shoulder seasons — Autumn demand steps down from 938 (no snow) → 530 → 371 → 275 (light → moderate → heavy snow), and Winter shows a similar (if noisier) decline.

Humidity:

Humidity group	Records	Avg. demand	Max demand
30-49%	2,488	840.37	3,418
50-69%	2,708	830.31	3,556
< 30%	630	719.02	3,365
70-89%	2,033	622.12	3,016
90%+	606	190.13	2,059

Demand peaks in the moderate 30-69% humidity range and falls off sharply above 90% (a **77% drop** relative to the 30-49% peak group) — muggy conditions substantially suppress ridership.

4.6 Holiday Effect

Metric	Value
Holiday hours	408 (4.8%)
Non-holiday hours	8,057 (95.2%)
Avg. rentals, holidays	529.15
Avg. rentals, non-holidays	739.29
Max rentals, holidays	2,400
Busiest hour, holidays	17:00 (2,400)
Busiest hour, overall/non-holidays	18:00 (3,556)

Holidays see **~28% lower average demand** than regular days, consistent with reduced commuting, and the holiday peak hour shifts an hour earlier (17:00 vs. 18:00) — plausibly reflecting leisure rather than commute-driven trips.

5. Global Bike-Sharing Systems Dataset

5.1 Overview

Metric	Value
Total records	480
Unique countries	57
Unique cities	434
Unique systems	87
Missing fleet size (BICYCLES = NA)	76 (15.8%)
Records with BICYCLES = 0	0

Data quality flag: 76 records (15.8%) are missing a fleet-size value entirely (NA).

To be addressed in the next DATA SCIENCE phase.

Duplicate CITY+SYSTEM combinations: 9 combinations appear more than once (e.g., Berlin/"4 Gen." appears 4 times; Berlin/"3 & 4 Gen. Call a Bike flex," Warsaw/"3 Gen. nextbike," and others appear twice). Three of these duplicate pairs involve a **missing system name (<NA>)** — e.g., Bratislava and Padua each have two records where the system name itself wasn't captured. **To be addressed in the next DATA SCIENCE phase.**

5.2 Fleet Size Distribution

Metric	Value
Minimum fleet size	5
Maximum fleet size	78,000
Average fleet size	2,012.2
Median fleet size	343

As with city population, fleet size is **heavily right-skewed** (mean $\sim 5.9\times$ the median) — a small number of massive municipal systems (e.g., an 78,000-bike system, consistent with large Chinese public-bike networks) dominate the average.

Query-logic flag: The query labeled “*Find the 10 largest bike-sharing systems by BICYCLES*” returns small values (113, 50, 26, 25, 23, 21...) that look like **counts of records per system name** (e.g., how many cities use “3 Gen. nextbike”), not summed fleet totals — compare this with the separate “*extremely large systems (>30,000 bikes)*” query below, whose totals are two to three orders of magnitude larger. These two outputs are inconsistent with each other and the first appears to use COUNT(*) where SUM(BICYCLES) was intended.

Systems with truly large (>30,000) total bicycle counts:

System	Total bicycles
(missing/<NA> system name)	461,601
PBSC & 8D	44,893
Forever Bicycle	42,715
3 Gen. Cyclocity	38,671

Data quality flag — significant: Records with a **missing system name (<NA>)** account for **461,601 bicycles**, by far the largest single “system” bucket in the entire dataset — larger than the next three named systems combined. This means a very large share of the world’s tracked bike-share capacity is currently **unattributed to any named operator. To be addressed in the next DATA SCIENCE phase.**

The 10 smallest systems by fleet size all show exactly **0 bicycles** (e.g., “Mobike,” “Zagster,” “Yellow Bikes,” “YoBike” — several of these are real bike-share brands that are known to have shut down or been discontinued, so a 0 here may reflect a genuinely retired system rather than a data error, but this should be confirmed rather than assumed).

5.3 Country-Level Analysis

Total bicycles by country (top 10 of 57, excluding NA):

Country	Total bicycles
China	469,941
France	49,552
United States	48,008
Germany	22,643
South Korea	22,348
Netherlands	22,250
Taiwan	19,567
Italy	17,998
United Kingdom	17,430
Canada	15,365

Across all countries with a known total (grand total $\approx$ **812,931 bicycles**), **China alone accounts for ~58%** of the world’s tracked bike-share fleet in this dataset — a dramatic concentration.

Data quality flag: Three countries — **Chile, Latvia, and Turkey** — have **NA** for both total and average fleet size, despite appearing in the country list (i.e., they have systems on record, just with missing bicycle counts). **To be addressed in the next DATA SCIENCE phase.**

Average fleet size per system, top 10 countries:

Country	Avg. fleet size
South Korea	11,174.0
China	8,102.4
Netherlands	7,416.7
Taiwan	6,522.3
Mexico	3,371.0
Canada	3,073.0
Iran	2,975.0
Belgium	2,638.3
Israel	2,000.0
Finland	1,755.0

Number of bike-share systems (i.e., records), top 10 countries:

Country	System count
Germany	59
United States	55
France	36
Greece	22
United Kingdom	15
Poland	15
Czechia	14
Switzerland	9
Spain	8
Brazil	8

This is a useful contrast to the fleet-size table above: **Germany and the U.S. lead in the number of systems** (broad, many small/medium deployments across many cities), while **China, South Korea, Netherlands, and Taiwan lead in fleet size per system** (fewer, much larger deployments). This reflects two different bike-share investment models — breadth vs. depth.

5.4 City-Level Analysis

21 cities have more than one bike-sharing system on record, led by **Berlin (9 separate system entries)** — likely reflecting successive/competing operators and system generations (labels like “3 Gen.” and “4 Gen.” suggest generational upgrades of the same underlying network rather than 9 concurrent independent systems) rather than genuinely 9 simultaneous fleets. Other multi-system cities include Düsseldorf, Frankfurt am Main, and Milan (4 each), and Bangkok, Nuremberg, Sydney (3 each).

Combined fleet totals for these multi-system cities range from **100 (Tübingen)** to **17,000 (Milan)**, with two cities — **Aschaffenburg and Freiburg im Breisgau** — **showing NA for their combined total** (i.e., missing bicycle counts even after combining all of that city’s system rows). **To be addressed in the next DATA SCIENCE phase.**

6. Cross-Dataset Join Analysis

6.1 Match Rate: World Cities ↔ Bike-Sharing Systems

Metric	Value
Bike-sharing cities matched to a world city	271 (62.4%)
Bike-sharing cities not matched	163 (37.6%)

Well over a third of bike-share cities cannot currently be linked to the world-cities gazetteer at all — almost certainly due to spelling/diacritic differences, alternate city names, or country-name mismatches. **This is a substantial gap that should be closed (e.g., via a fuzzy-match or alias table) before population-normalized bike-share metrics are trusted at scale.**

6.2 The Join Fan-Out Problem

Of the 271 cities that *do* match, **57 produce more than one matching row** — meaning the join is not being keyed on a unique identifier (like CITY + COUNTRY), but appears to match on **city name alone**, which is not unique globally.

Worst-affected cities (number of world-city rows matched):

City	Matches
Berlin	36
Milan	12
Hamilton	11
Victoria	9
Vienna	8
Florence	8
Santiago	7
Paris	6
Warsaw	4
Valencia	4

Concrete worked example — why this matters: One query, “*Find the largest cities with weather forecasts,*” reports Paris’s average population as **1,847,093**. Tracing the underlying numbers shows this is the straight average of Paris’s real population (11,020,000, France) blended with **five much smaller places also named “Paris”** in the world-cities table (populations of 5,131 / 8,394 / 10,536 / 12,485 / 26,010 — almost certainly small U.S. towns, e.g. Paris, Texas): $(5,131 + 8,394 + 10,536 + 12,485 + 26,010 + 11,020,000) \div 6 = 1,847,093$, exactly matching the reported figure. **The real Paris, France has a population of 11 million — this join has diluted it down to under 2 million by blending in unrelated small towns.**

By contrast, a separate query (“*Compare population with temperature, humidity and wind speed*”) reports Paris’s population correctly as **11,020,000** — showing that **different queries in this same dump handle the same ambiguous join inconsistently**, sometimes correctly isolating the right city and sometimes averaging across all name-alike matches. **Every downstream number that depends on this join should be re-derived using a CITY+COUNTRY (or better, a stable city ID) key before being used for decision-making.**

A second, distinct cause of fan-out also appears: **Suzhou, China** is a legitimate single city with one bike-sharing fleet (22,940 bikes), but because `WORLD_CITIES` itself contains **two duplicate Suzhou, China records** (populations 5,250,000 and 5,690,000 — see Section 2.2), the join produces two rows for Suzhou, both carrying the same fleet size. This is a knock-on effect of the base-table duplication issue, not a naming collision.

6.3 Bicycles-per-1,000-Population — Usable vs. Unusable Results

A large (349-row) table of bicycles-per-1,000-population is present in the dump. Because of the fan-out issue above, its **top entries are not meaningful**: e.g., “Paris” tops the list at **2,825.96 bicycles per 1,000 residents**, which is derived from dividing central Paris’s 14,500-bike fleet by the population of a small same-named U.S. town (5,131) — not a real per-capita figure for any actual place. Similar artifacts inflate entries for “Milan,” “Berlin,” and others near the top of that ranking.

A separate, better-controlled pair of queries (“*cities with large populations but relatively small fleets*” / “*...small populations but relatively large fleets*”) appears to key on **CITY + COUNTRY** and produces far more trustworthy results:

Selected large megacities — bikes per 1,000 population (low end, as expected for huge metro populations):

City	Country	Bicycles	Population	Bikes / 1,000
Shanghai	China	19,165	22,120,000	0.87
São Paulo	Brazil	600	22,046,000	0.03
Mexico City	Mexico	6,500	20,996,000	0.31
Guangzhou	China	4,840	20,902,000	0.23
Beijing	China	16,000	19,433,000	0.82
Paris	France	14,500	11,020,000	1.32

Even with the correct pairing, three flagged “small population, large fleet” outliers remain suspicious and are very likely **more fan-out artifacts, not genuine outliers**:

City	Country	Bicycles	Population	Bikes / 1,000
Atlanta	United States	500	5,526	90.48
Lanxi	China	1,500	72,528	20.68
Didymóteicho	Greece	100	9,263	10.80

For Atlanta specifically, the correctly-matched record elsewhere in the dump shows the real Atlanta, GA (population 5,449,398) with the same 500-bike fleet, giving a far more plausible 0.09 per 1,000 — strongly suggesting the 90.48 figure above comes from a different, much smaller place also called “Atlanta.” **Recommend re-running all per-capita bike-share metrics on a corrected join before relying on them.**

6.4 City Case Studies: Seoul, Paris, Suzhou (and the Washington, D.C. Gap)

Only **3 of the 4** weather-forecast cities (Paris, Seoul, Suzhou) successfully match a bike-share system record; **Washington, D.C. does not**, despite having a well-known public bike-share program in reality — this is a strong candidate for a naming mismatch (e.g., “Washington” vs. “Washington, D.C.”) rather than a genuine absence, and is grouped with the 163 non-matching cities from Section 6.1.

City	Fleet size	Avg. Temp (°C)	Avg. Humidity (%)	Avg. Wind Speed
Suzhou	22,940	23.68	71.10	2.31
Seoul	20,000	22.71	44.53	1.91
Paris	14,500	19.55	51.95	2.61

Cutting this further by “favorable” vs. “unfavorable” forecast windows shows, unsurprisingly, that the same three large-fleet cities appear in both groups — the classification is really separating individual 3-hourly readings by their conditions (e.g., high humidity/wind for Suzhou readings in the 80–95% humidity range vs. more comfortable 60–70% readings), rather than distinguishing between cities. This is a reasonable way to explore *when* conditions are favorable for a given city’s fleet, but should not be read as ranking cities against each other.

6.5 Final Targeted Queries

- An explicit join to find Seoul’s total bike count alongside its city attributes correctly returns **20,000 bicycles, population 21,794,000, coordinates (37.58, 127.00)**.
- A query for “**all cities with total bike counts between 15,000 and 20,000**” returns 9 rows — and again illustrates the fan-out problem directly: “**Milan**” **appears three times**, once correctly (Italy, population 1,351,562, 17,000 bikes) and twice more attached to two small U.S. towns also named Milan (populations 7,201 and 7,778) that inherit the same Italian fleet size of 17,000 bikes. This is the same root cause as Section 6.2, shown one more time in a fresh query.

7. Consolidated Data-Quality Log

For quick reference, every NULL/NA and notable anomaly identified in this pass:

#	Dataset	Issue	Scale	Status
1	World Cities	Missing population (NA)	973 rows (3.7%)	→ Data Science phase
2	World Cities	ISO2 code count ≠ country/ISO3 count	223 vs. 224	→ Data Science phase
3	World Cities	Duplicate CITY+COUNTRY combinations	1,073 combos / ~1,881 extra rows (7.1%)	→ Data Science phase
4	World Cities	Implausible tiny populations (e.g., “4.472”; several exact “10”s)	≥10 rows	Flagged for review
5	World Cities	“Largest city per country” excludes 3 countries	221 of 224	Likely caused by #1
6	World Cities	“Sort by highest population” query returns unsorted output	1 query	Query-logic issue

#	Dataset	Issue	Scale	Status
7	Weather Forecast	All records mislabeled	160/160 rows	Query-logic issue
8	Weather Forecast	“Winter” season “Next 3 hours” query returns 1 row with raw epoch timestamp	1 query	Query-logic issue
9	Weather Forecast	“Largest city per weather condition” returns malformed labels	1 query	Query-logic issue
10	Weather Forecast	“Largest temperature amplitude” inconsistent with actual max–min range	1 query	Query-logic issue
11	Seoul Bike-Sharing	12 calendar dates missing entirely (~295 missing hours)	12 of 365 days	→ Data Science phase
12	Seoul Bike-Sharing	“Busiest/quietest date” and “busiest/quietest hour per season” return single extreme records, not true aggregates	2 queries	Query-logic issue
13	Seoul Bike-Sharing	Snow-bucket averages not perfectly monotonic (small samples)	2 buckets	Flagged for review
14	Bike-Sharing Systems	Missing fleet size (NA)	76 records (15.8%)	→ Data Science phase
15	Bike-Sharing Systems	3 countries (Chile, Latvia, Turkey) with NA total/average fleet	3 countries	→ Data Science phase
16	Bike-Sharing Systems	2 cities (Aschaffenburg, Freiburg im Breisgau) NA combined fleet	2 cities	→ Data Science phase

#	Dataset	Issue	Scale	Status
17	Bike-Sharing Systems	Duplicate CITY+SYSTEM rows, some with missing (<NA>) system name	9 combos	→ Data Science phase
18	Bike-Sharing Systems	“Largest systems by BICYCLES” appears to count records, not sum fleet	1 query	Query-logic issue
19	Bike-Sharing Systems	461,601 bicycles (the single largest bucket) attributed to a missing (<NA>) system name	1 bucket	→ Data Science phase
20	Cross-join	163 of 434 bike-share cities fail to match WORLD_CITIES	37.6%	Join needs rework
21	Cross-join	57 cities match multiple WORLD_CITIES rows (name-only join fan-out)	57 cities	Join needs rework
22	Cross-join	Paris average population distorted to 1.85M by fan-out (true: 11.02M)	1 worked example	Join needs rework
23	Cross-join	Bicycles-per-1,000 top results (Paris, Milan, etc.) are join artifacts	Top of 349-row table	Join needs rework
24	Cross-join	Washington, D.C. fails to match any bike-share system despite having one in reality	1 city	Join needs rework

8. Recommendations for the Data Science Phase

1. **Resolve all NA/NULL values** logged in Section 7 (population, fleet size, ISO codes, country-level fleet totals) — through source re-acquisition, imputation, or explicit exclu-

sion, as appropriate per field.

2. **De-duplicate** WORLD_CITIES on a proper key (e.g., CITY + COUNTRY + coordinates) before it is used in any further joins — this alone would fix the Suzhou-style fan-out and likely reduce the U.S.-driven population inflation seen in Section 2.5.
3. **Rebuild the** WORLD_CITIES ↔ BIKE_SHARING_SYSTEMS **join on CITY + COUNTRY (or a shared city ID)**, not city name alone — this is the single highest-impact fix, as it corrupts every per-capita and cross-dataset comparison in Section 6.
4. **Investigate and re-run the ~9 flagged “query-logic” issues** in Section 7 (the unsorted “highest population” query, the mislabeled weather seasons, the malformed “largest city per weather condition,” the count-vs-sum confusion in bike-share system rankings, and the single-record “busiest date/hour” queries) — none of these require new data, just corrected SQL.
5. **Confirm whether “0 bicycle” systems (e.g., Mobike, Zagster, YoBike) are genuinely defunct** rather than missing data, and label them accordingly rather than leaving them ambiguous.
6. **Standardize city-name matching** (accents, “D.C.” style suffixes, alternate spellings) to close the 37.6% non-match rate between the two city-level datasets.

End of report.